PRE-BOARD EXAMINATION (2024-25)
CLASS X

TIME: 3 HOURS **SUBJECT-SCIENCE (086)** **MM: 80**

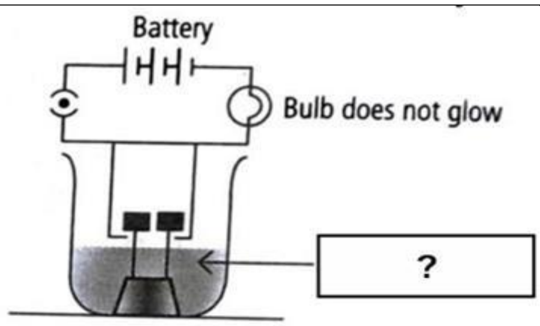
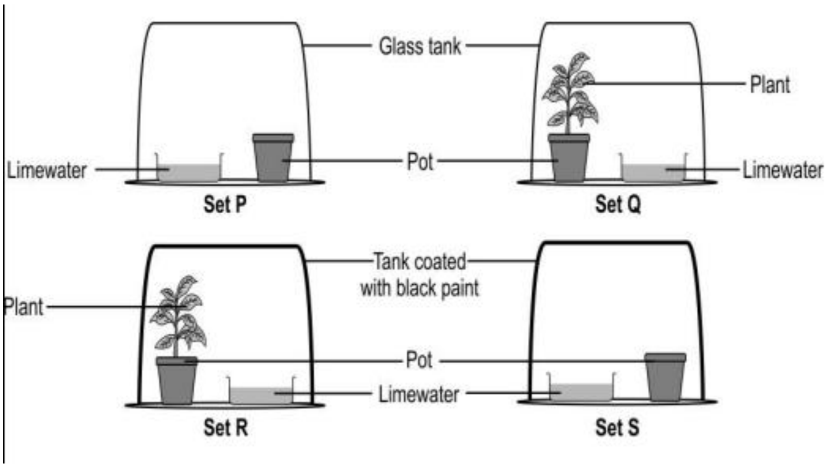
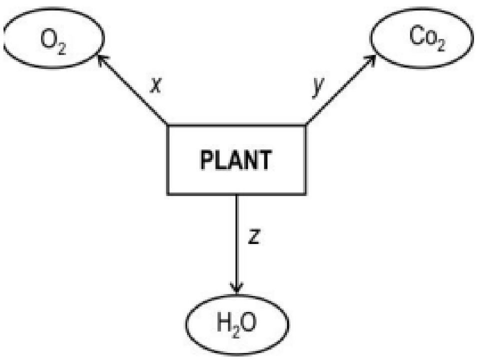
General Instructions:

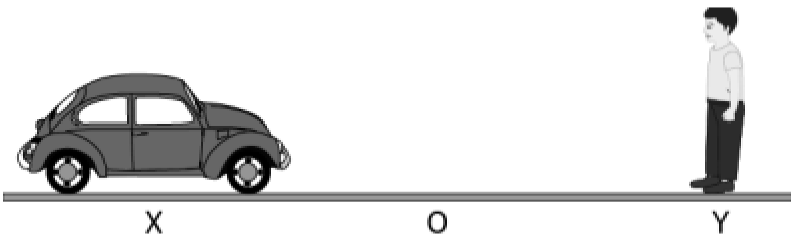
1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

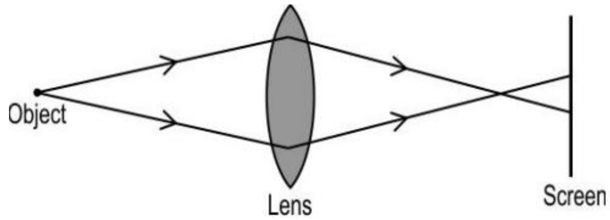
SECTION- A

Select and write one most appropriate option out of the four options given for each of the question 1-20.

1.	In the following case(s) the combination reaction occurs in : I. $CuO + H_2 \Rightarrow$ II. $ZnO + C \Rightarrow$ III. $Na + O_2 \Rightarrow$ IV. $CH_4 + O_2 \Rightarrow$ (A) Only III (B) Only IV (C) II & III (D) I, III & IV	**1**
2.	When 50g of lead powder is added to 300 ml of blue copper sulphate solution, after a few hours, the solution becomes colourless. This is an example of A. Combination reaction B. Decomposition reaction C. Displacement reaction D. Double displacement reaction	**1**
3.	Which of the following is an endothermic reaction? A. Burning of candle. B. Cooking of food. C. Decomposition of vegetable matter. D. Reaction of sodium with air	**1**
4.	A substance X, turns red litmus blue, it will change methyl orange to: A. Yellow B. Pink C. Red D. Colourless	**1**
5.	Match column I with column II and select the correct option using the given codes a. A metal that forms amphoteric oxides — (i) Ga b. A metal which melts when kept on our palm — (ii) Au c. A metal that reacts with nitric acid — (iii) Al d. A metal which cannot displace hydrogen from acids — (iv) Mn A. a – (ii), b – (i), c – (iii), d – (iv) B. a – (iii), b – (i), c – (iv), d – (ii) C. a – (iv), b – (ii), c – (iii), d – (i) D. a – (iii), b – (ii), c – (i), d – (iv)	**1**

6.	In snails individuals can begin life as male and depending on environmental conditions they can become female as they grow. This is because A. male snails have dominant genetic makeup. B. female snails have dominant genetic makeup. C. expression of sex chromosomes can change in a snail's life time. D. sex is not genetically determined in snails.	**1**
7.	 The solution in the given figure is likely to be A. HNO_3 B. C_2H_5OH C. H_2SO_4 D. CO_2 in water	**1**
8.	Lime water turns cloudy in the presence of a gas which is a by-product of respiration. Shown below are four setups kept in sunlight for 24 hours. In which setup is lime water expected to be the cloudiest?  A. Setup P B. Setup Q C. Setup R D. Setup S	**1**
9.	Look at the diagram below carefully. Identify the process taking place at Z.  A. Reproduction B. Transpiration C. Photosynthesis D. Translocation	**1**
10.	Reaction between X and Y, forms compound Z. X loses electron and Y gains electron. Which of the following properties is not shown by Z? A. Has high melting point B. Has low melting point C. Conducts electricity in molten state D. Occurs as solid	**1**

11.	Which of the following statements is/are true about a neuron ? I. Dendrites of neuron pass the impulse to the axon. II. Axon of neuron carries the impulse from the cell body. III. Sensory neuron carries the impulse to the specific effectors. IV. Transmission of impulse from a neuron to a muscle fibre occurs through neuro-muscular junction. (A) I only (B) I and III (C) II and III (D) II and IV	**1**
12.	The opening and closing of stomatal pore depends upon: A) temperature B) excess of CO_2 in the guard cells. C) high concentration of oxygen in the guard cells. D) flow of water in and out of the guard cells.	**1**
13.	In the following cases, a ray is incident on a concave mirror. In which case is the angle of incidence equal to zero? A. A ray parallel to the principal axis. B. A ray passing through the centre of curvature and incident obliquely. C. A ray passing through the principal focus and incident obliquely. D. A ray incident obliquely to the principal axis, at the pole of the mirror.	**1**
14.	A person standing at point Y is watching a car coming from a point X to O as shown  1. at X the focal length is higher than at O 2. at O the focal length is higher than at X 3. at X the ciliary muscle is thicker than at O 4. at O the ciliary muscle is thicker than at X Which change in the person's eye would likely to occur while watching the car? A. 1 and 3 B. 1 and 4 C. 2 and 3 D. 2 and 4	**1**
15.	In 1987, an agreement was formulated by the United Nations Environment Programme (UNEP) to freeze the production of "X" to prevent depletion of "Y". "X" and "Y" respectively referred here are: A. Ozone; CFCs B. CFCs; rays UV C. CFCs; Ozone D. UV rays; Diatomic oxygen	**1**
16.	Which of the following is not a role of decomposers in the ecosystem? A. Natural replenishment of soil. B. Enrichment of oxygen in atmosphere. C. Waste decomposition. D. Break-down of dead remains.	**1**

	Q. No. 17 to 20 are Assertion - Reasoning based questions. These consist of two statements –Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below- A. Both A and R are true and R is the correct explanation of A B. Both A and R are true and R is not the correct explanation of A C. A is true but R is false D. A is false but R is true	
17.	**Assertion (A):** If the lions are removed from a food chain it will not affect the food chain, however if the plants are removed from a food chain it will disturb the ecosystem. **Reason (R)**: Plants are producers who can make food using sunlight, while lions are consumers.	**1**
18.	**Assertion (A):** Probability of survival of an organism produced through sexual reproduction is more than that of organism produced through asexual mode. **Reason (R):** Variations provide advantages to individuals for survival.	**1**
19.	**Assertion (A):** A convex lens can form a magnified erect as well as magnified inverted image of an object placed in front of it. **Reason (R):** A magnified and an inverted image can be obtained by a convex lens when an object is kept between F and O.	**1**
20.	**Assertion (A):** On adding dil. HCl to a test tube containing a substance 'X', a colourless gas is produced which gives a pop sound when a burning match stick is brought near it. **Reason (R):** In this reaction metal 'X' is displaced by Hydrogen.	**1**
	SECTION- B **Q. no. 21 to 26 are very short answer questions: (VSA)**	
21.	What is the purpose of making urine in the human body? Name the organs that stores and releases the urine.	**2**
22.	Identify the type of each of the following reactions stating the reason for your answers. A. $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe + heat$ B. $Pb(NO_3)_2 + 2KI \rightarrow PbI_2(\downarrow) + 2KNO_3$	**2**
23.	***<u>Attempt either option A or B.</u>*** A. Explain the role of the following enzymes in the process of digestion of food in humans: (i) Salivary amylase (ii) Pepsin (iii) Trypsin (iv) Lipase **OR** B.(a) Oxygen gas is released out during photosynthesis. Which raw material of photosynthesis is the source of oxygen gas? (b) What are the steps involved in photosynthesis after absorption of sunlight by chlorophyll? (c) What energy transformation takes place in photosynthesis?	**2**
24.	(a) A lens forms a blurred image of an object on the screen as shown below:  What changes can you make to the following to form a sharp and in-focus image on the screen? (i) Object distance (ii) Focal length of the lens	**2**

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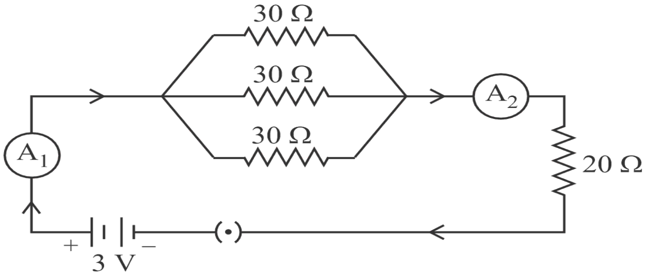
<u>Attempt either option A or B</u>

A. On the basis of the given circuit answer the questions given below :



i Write the relationship between the readings of the two ammeters A1 and A2.

ii. Give reason for your answer.

OR

B. You are given 2 fuse wires A and B with current ratings 2A and 5A respectively. Justify with reason which of the two would you use with a 1000W, 220V room heater?

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<tr><td>26.</td><td>

(i) What is the name given to organisms belonging to first and third trophic level?

(ii) Differentiate them on the basis of their feeding habit.

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SECTION – C

Q. No. 27 to 33 are short answer questions (SA)

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A. State the relationship between the resistance R of a wire to its length l and cross- sectional area A. Use the mathematical symbols to arrive at the final formula.

B. Using the formula define the resistivity of a material.

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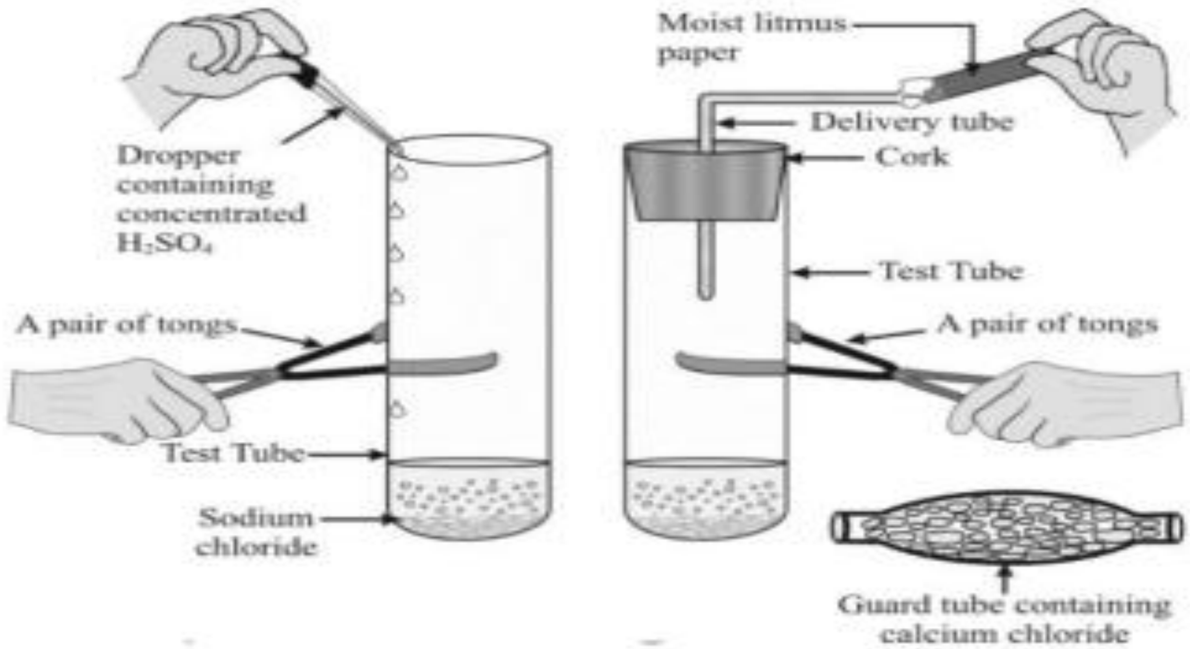
<tr><td>28</td><td>

<u>***Attempt either option A or B***</u>.

A.

Observe the given figure and answer the following questions:



(a) The above image represents preparation of gas X. Identify X.

(b) Dry X gas does not change the colour of the dry blue litmus paper . Give reasons

(c) It is suggested that if weather is humid, the gas need to passed through guard tube containing calcium chloride. Why is it so?

OR

B

(i) In the given series of reactions, name the compounds X and Z.

(ii) Which type of reaction is X to Z?

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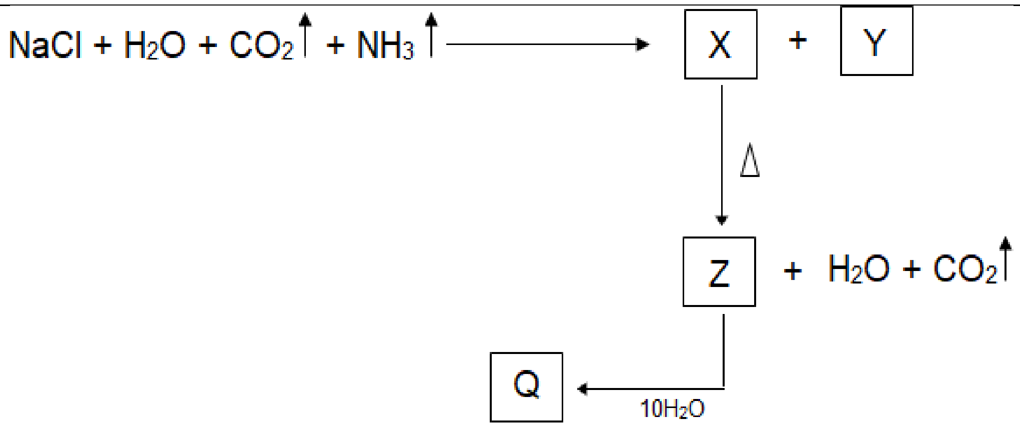
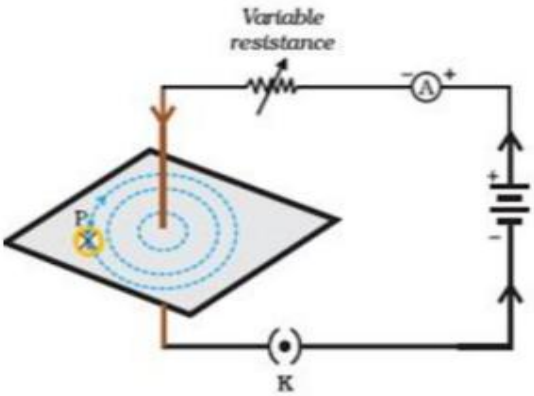
	 (iii)You are given 3 unknown solutions A, B, and C with pH values of 6, 8 and 9.5 respectively. In which solution will the maximum number of hydronium ions be present? (iv)Arrange the given samples in the increasing order of H^+ ion concentration.	
29.	A. How is the brain protected from shocks and injuries? B. Which part of the brain is involved in activities like (i) picking a pencil and (ii) vomiting? Also state whether these actions are voluntary or involuntary.	**3**
30.	A. In a family of four individuals, the father possessed long ears and the mother possessed short ears. If the parents had pure dominant and recessive traits respectively, then calculate the ratio of genetic makeup of F2 generation. Show a suitable cross. B. If father had short ears and the mother had long ears, explain what effect it will have on the ratio of genetic makeup in F2 generation.	**3**
31.	Sunita's ophthalmologist suggests her to use a lens of power -2 D to correct an eye defect. (i) Can she use a convex lens? Why? (ii) What should be the focal length of the lens? (iii) An object is kept at 10 cm in front of the lens of power -2 D. Find the distance where the image is produced.	**3**
32.	Metal X is found in nature as its sulphide XS. It is used in the galvanisation of iron articles. Identify the metal X. How will you convert this sulphide ore into the metal? Explain with equations.	**3**
33.	 In the activity with a magnetic compass and a straight current-carrying wire, it is observed that as the compass is moved away from the current-carrying wire, the deflection of the compass needle reduced. A. Explain why the deflection of the compass needle reduces as it is moved away from the current carrying wire. B. Mention one thing that can be changed in the circuit of the wire that could increase the deflection of the needle. C. Explain with reason how the direction of the magnetic field associated with the wire changes if the polarity of battery reversed.	**3**

SECTION- D
Q.no.34 to 36 are long answer questions (LA)

34. ***Attempt either option A or B.*** **5**

A. A saturated organic compound ‘A’ belongs to the homologous series of alcohols.
On heating A with concentrated sulphuric acid at 443 K, it forms an unsaturated compound B with molecular mass 28u.
The compound B on addition of one one mole of hydrogen in the presence of Nickel, changes to a saturated hydrocarbon C.
i. Identify A, B, C
ii. Write the chemical equation showing the conversion of A into B.
iii. What happens when compound C undergoes combustion?
iv. State one industrial application of hydrogenation reaction.
v. Name the product formed when compound A reacts with sodium.

OR

B.
(i) ‘A’ & ‘B’ are sodium salts of long-chain carboxylic acid and long chain Sulphonic acid respectively. What names are given to A & B?
(ii) With the help of diagram, show the formation of micelles, when soap is applied on oily dirt.
(iii) Which among A or B forms curdy white precipitate with hard water? Why?
(iv) Write the chemical equation of the preparation of soap from an ester. What is the name of this process?

35. **Attempt either option A or B.** **5**

A. Plants are propagated by sexual or asexual means in fields and nurseries. Over the years horticulturists have developed asexual methods that use vegetative parts of the plants to multiply. Many plants can reproduce by this method naturally as well as by artificial means.

(i) Give one example of (a) a flower, and (b) a fruit grown by vegetative propagation.
(ii) List two advantages the farmer will have, using vegetative propogation.
(iii) List 2 reasons why is the offspring of this Bryophyllum plant may not absolutely identical to its parent plant?

OR

B. (i) In human females, 1 egg is matured and released from either of the ovary each month. What happens to the egg -
a. when it is fertilized?
b. when it is not fertilized?
(ii) Name the region where the embryo develops inside the female body. Explain how this region is adapted for nourishing the baby.
(iii) How does exchange of substances take place between the foetus and mother’s womb?

36. **Attempt either option A or B** **5**

A
(i) State the law and write the formula connecting the electric current flowing through a conductor and voltage applied across it.
(ii) In a household five bulbs each of 200W are used for 5 hours and an electric press of 1000W for 2 hours every day. Calculate the cost of using the fans and electric press for 60 days if the cost of 1 unit of electrical energy is Rs. 7.5.

OR

B. (i) State Joules' law of heating and write its mathematical expression.

(ii) Two resistors of resistances 2 ohm and 4 ohm are connected in

a) series
b) parallel

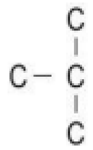
with a battery of given potential difference. Compute the ratio of total quantity of heat produced in the combination in the two cases if the total voltage and time are kept the same for both.

SECTION-E

Q.no.37 to 39 are case-based / data-based questions.

37. **Attempt either of subpart C or D** — 4

Given below is a four carbon skeleton of a hydrocarbon compound.



Fill in the hydrogen atoms/bonds to form:

A. a saturated hydrocarbon

B. an unsaturated hydrocarbon

Student to attempt either subpart C or D

C. If the four-carbon skeleton is of a straight chained alkene, draw the structures of all the possible compounds.

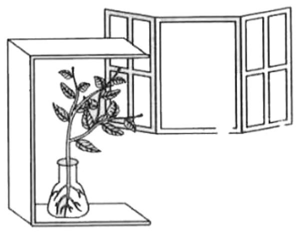
OR

D. If the four-carbon skeleton is of a straight chained alkyne:

(i) How many carbon atoms may NOT be bonded to any hydrogen atoms?

(ii) How many hydrogen atoms will there be in the compound?

38. **Observe the given pictures to answer the following questions.** — 4



A. Shoots of a plant bending towards light

B. Folding of leaves Touch me not plant

Attempt either subpart A or B

A. What causes the bending of shoots in the plants as shown in figure A?

OR

B. What causes the folding of the leaves in 'Touch me not' plant as shown in figure B?

C. Compare the movement of growth of the pollen tube towards ovule with the movements shown in part A of the above figure.

D. Compare the movement shown in figure B with the movement of body parts in the animals.

39.	In our laboratories we use compound microscopes to see the magnified image of a microscopic object. A compound microscope is made up of two lenses. The lens nearest to the object to be viewed, called objective lens, forms real, inverted and magnified image of the object. This image serves as an object for the second lens called eyepiece. The eyepiece forms virtual, erect and magnified image of its object. Thus, the resultant image formed by a microscope is virtual, inverted and magnified with respect to the microscopic object viewed. A. What kind of image is seen through a compound microscope? B. The image of an object formed by a convex lens of focal length 6 cm is virtual, erect and magnified. What is the range of object distance in this case? **Attempt either subpart C or D** C. An object is placed at a distance of 12 cm from the optical centre of a convex lens of focal length 18 cm. Draw a labelled ray diagram to show the formation of image in this case. **OR** D. The image formed by a convex lens is real, inverted and of the same size as the object. If the distance between the objects and the image is 60 cm, determine the focal length of the lens. Give justification for your answer.	**4**